

	Since y is vertical displacement, at maximum height (largest y value) $E = 0$. Hence the answer is A.
6	Ans: B When the motor is not spinning, the 60 N mass will move downwards as there is a net downward force of 35 N. Since the belt remains stationary when the motor is spinning, $P_{\text{output}} = F_{\text{net}} \times (\text{distance moved per unit time}) = F_{\text{net}} \times (20\pi d)$ $= 35 \times (20 \times \pi \times 0.35) = 770 \text{ W}$
7	Ans: A